Software Requirements Specification

SmartStock — Supermarket Inventory,

Sales Prediction & Payroll Management System

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Chapter 1

Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for SmartStock, a system for managing supermarket inventory, forecasting product demand from sales history, administering staff and payroll, and processing customer payments through mobile money. It is derived from the approved Software Design Document (SDD) for the system and restates the system's behaviour as verifiable requirements, independent of the specific design and implementation choices described in the SDD. It is intended to serve as an agreed basis between the development team and course evaluators for what the system must do.

1.2 Scope

SmartStock is a supermarket management platform covering four functional areas:

- **Inventory management** – tracking products, stock levels, suppliers, and reorder points.
- **Demand prediction** – forecasting, from historical sales, which products will be needed more or less over time and in what quantity.
- **Staff and payroll management** – maintaining staff records and roles, and processing salaries.
- **Payment processing** – accepting customer payments and disbursing staff payouts through mobile money (e.g., MTN MoMo, Orange Money).

The system serves three internal actors – Customer, Cashier, and Manager/Admin – and integrates with an external Mobile Money Gateway (and, where applicable, an Online Payment Gateway) for payment authorization.

The system does *not* cover: physical point-of-sale hardware drivers, tax-authority filing/reporting integrations, or multi-store/warehouse logistics beyond a single supermarket's stock, unless added in a later revision of this document.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
UML	Unified Modeling Language
MoMo	Mobile Money (e.g. MTN MoMo, Orange Money)
POS	Point of Sale
SKU	Stock Keeping Unit
Forecast	A system-generated prediction of future demand for a product

Term	Definition
SRS	Software Requirements Specification
SDD	Software Design Document
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 References

- SmartStock Software Design Document, Version 1.0, Group 2, CS 4122, ICT University Yaóundé.
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications (structural reference for this document).

1.5 Document Overview

Chapter 2 gives an overall description of the product, its functions, users, and constraints. Chapter 3 presents specific requirements: functional requirements organized by functional area, external interface requirements, and use-case-driven behaviour. Chapter 4 presents non-functional requirements. Chapter 5 lists data requirements derived from the domain model. Chapter 6 records assumptions, dependencies, and traceability to the SDD.

Chapter 2

Overall Description

2.1 Product Perspective

SmartStock is a new, self-contained supermarket management system composed of cooperating logical components: a Point-of-Sale (POS) front end, an Inventory Service, a Payment Service, a Prediction Engine, a Notification Service, and a Payroll subsystem. The Prediction Engine is deliberately decoupled from core inventory logic: it consumes historical sales data and returns forecasts, allowing the forecasting model to evolve independently of transactional processing. The system exchanges data with one primary external system, the Mobile Money Gateway, for both customer payment authorization and staff payroll payouts, and optionally an Online Payment Gateway for card-based payments.

2.2 Product Functions

At a high level, SmartStock shall allow the system to:

1. Register and authenticate Customers, Cashiers, and Managers/Admins.
2. Let Customers browse products, view order history, and check out.
3. Let Cashiers scan/select items, compute totals, and process payments (mobile money or online payment).
4. Let Managers/Admins manage inventory, suppliers, staff records, payroll, and view sales/stock reports.
5. Generate demand forecasts per product from historical sales data and flag replenishment candidates.
6. Track the stock-status lifecycle of inventory items and drive replenishment workflows.
7. Compute and disburse staff payroll via the Mobile Money Gateway.

2.3 User Classes and Characteristics

User Class	Characteristics
Customer	End user who registers, browses/searches products, views order history, and checks out and pays via mobile money or online payment. Assumed to have basic digital literacy and a registered mobile money account for MoMo payments.

User Class	Characteristics
Cashier	Operates the POS at checkout: authenticates into the system, scans/selects items for a sale, selects a payment method, and completes or retries a payment. Assumed to be a trained employee with routine daily use of the system.
Manager/Admin	Highest-privilege internal user. Manages inventory, suppliers, staff records, payroll runs, and demand forecasts, and views sales/stock reports. Assumed to have elevated authorization and periodic (not necessarily daily) use.
Mobile Money Gateway (external)	Not a human user but an external system actor that authorizes customer payments and executes staff payouts on request.

2.4 Operating Environment

SmartStock is a distributed, service-oriented system intended to run across POS terminals (cashier-facing), a customer-facing interface, and a manager back office, all communicating with backend services (Inventory Service, Payment Service, Prediction Engine, Notification Service, Payroll) over a network connection. It integrates with external mobile money and (optionally) online payment gateways over secure network channels. Specific deployment technology (cloud provider, containers, database engine) is a design decision documented in the SDD and is out of scope for this requirements-level document.

2.5 Design and Implementation Constraints

- Payments must be reconciled against a mobile money reference (`momoReference`) returned by the gateway; stock must not be decremented before payment is approved.
- Staff authentication must reuse a common credential/role model shared across all staff-facing actors (Cashier, Manager/Admin).
- Demand forecasting must be logically separable from transactional (sales/inventory) processing so that forecast computation does not block point-of-sale operations.
- The system must support at least two payment channels: mobile money and, optionally, online (card) payment, both mediated through an external gateway.

2.6 Assumptions and Dependencies

- The Mobile Money Gateway (and Online Payment Gateway, if used) is available, correctly integrated, and can approve, decline, or time out a payment request.
- Customers hold an active mobile money account prior to attempting a MoMo payment.
- Historical sales data of sufficient volume and quality exists for the Prediction Engine to generate meaningful forecasts.
- Suppliers can receive and fulfil purchase orders raised by the system within a communicated lead time.

Chapter 3

Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- A customer-facing interface for registration, login, product browsing/search, order history, and checkout.
- A cashier-facing POS interface for login, item scanning/selection, payment method selection, and receipt printing.
- A manager/admin back-office interface for inventory, supplier, staff, payroll, and reporting management.

3.1.2 Hardware Interfaces

The POS interface shall support item input (e.g., barcode scanning or manual product selection) and receipt output (printed or digital receipt) at the cashier station.

3.1.3 Software Interfaces

Interface	Description
Mobile Money Gateway	Receives payment requests (amount, phone number) and payout requests (staff payroll); returns approval, decline, or timeout, plus a transaction/momo reference.
Online Payment Gateway	Receives checkout-session requests for card payments; returns a hosted payment page redirect and an authorization result.
Sales History Repository	Supplies historical sales records to the Prediction Engine on request.

3.1.4 Communications Interfaces

All communication between the POS/customer/manager interfaces and backend services, and between backend services and external payment gateways, shall occur over a network using request/response messaging, with support for asynchronous/timeout handling on gateway calls.

3.2 Functional Requirements

3.2.1 Account and Authentication

ID	Requirement
FR-1.1	The system shall allow a Customer to register/create an account.
FR-1.2	The system shall allow Customers, Cashiers, and Managers/Admins to log in and authenticate using a username and password.
FR-1.3	The system shall allow an authenticated user to log out.
FR-1.4	The system shall associate each authenticated account with a role (Customer, Cashier, or Manager/Admin) that determines the actions available to it.

3.2.2 Customer and Sales / Checkout

ID	Requirement
FR-2.1	The system shall allow a Customer to browse and search products.
FR-2.2	The system shall allow a Customer to view their order history.
FR-2.3	The system shall allow a Cashier to scan or select items for a sale and compute the sale total.
FR-2.4	The system shall allow a Cashier to select a payment method (mobile money or online payment) for a checkout.
FR-2.5	The system shall check product stock availability before allowing a sale to proceed.
FR-2.6	The system shall print or issue a receipt once a payment is confirmed.
FR-2.7	The system shall allow a sale to be cancelled prior to payment confirmation.

3.2.3 Payment Processing

ID	Requirement
FR-3.1	When mobile money is selected, the system shall send a payment request (amount, phone number) to the Mobile Money Gateway.
FR-3.2	The system shall wait for the gateway to prompt the customer for PIN confirmation and shall receive an approval, decline, or timeout response.
FR-3.3	When online payment is selected, the system shall create a checkout session and redirect the customer to a hosted payment page for card details, then receive an authorization result.
FR-3.4	If a payment is declined or times out, the system shall prompt the Cashier to retry the same method or select an alternate payment method.
FR-3.5	The system shall not decrement stock or mark a sale as paid until payment approval has been received.
FR-3.6	The system shall store a payment record linked one-to-one with its sale, including the mobile money reference or transaction reference returned by the gateway, to support later reconciliation.

3.2.4 Inventory Management

ID	Requirement
FR-4.1	The system shall allow a Manager/Admin to add, update, and remove products, including price, category, and reorder level.
FR-4.2	The system shall allow a Manager/Admin to manage supplier records, including contact details and lead time.
FR-4.3	The system shall update stock quantities automatically whenever a sale is completed or new stock is received.
FR-4.4	The system shall determine whether a product is low on stock based on its configured reorder level.
FR-4.5	The system shall track individual inventory batches, including quantity, expiry date, and date received, and shall determine whether a batch is expired.
FR-4.6	The system shall maintain a stock-status lifecycle for each inventory item (In Stock, Low Stock, Out of Stock, Reordered, Discontinued) and transition an item's status automatically based on triggering events (e.g., sales reducing quantity, quantity reaching zero, purchase order placement, new stock received, or the item being delisted).

3.2.5 Demand Prediction

ID	Requirement
FR-5.1	The system shall allow a Manager/Admin to request a demand forecast for a given period.
FR-5.2	For each product in the catalog, the system shall fetch historical sales data for the requested period and generate a predicted quantity and trend using the Prediction Engine.
FR-5.3	The system shall store generated forecasts, associated with the product and forecast period.
FR-5.4	If any product's predicted demand falls below its reorder level, the system shall flag it as a replenishment candidate and notify the Manager/Admin.
FR-5.5	The system shall display forecast results per product to the Manager/Admin.

3.2.6 Stock Replenishment

ID	Requirement
FR-6.1	The system shall compare current stock against predicted need for each product and, when stock is insufficient, generate a purchase order to the relevant supplier.
FR-6.2	The system shall allow received stock to be verified against the original purchase order upon delivery.
FR-6.3	If a delivery matches the purchase order, the system shall update inventory levels accordingly; if it does not match, the system shall flag the discrepancy for Manager/Admin review.

3.2.7 Staff and Payroll Management

ID	Requirement
FR-7.1	The system shall allow a Manager/Admin to manage staff records, including role and base salary.
FR-7.2	The system shall allow staff to clock in.
FR-7.3	The system shall allow a Manager/Admin to run payroll for a given period, retrieving the list of staff and their role/salary data.
FR-7.4	For each staff member, the system shall compute net pay as gross pay minus deductions (e.g., tax, social security).
FR-7.5	The system shall send computed payouts to staff via the Mobile Money Gateway.
FR-7.6	If one or more payouts fail, the system shall record the partial failure and mark the affected staff members for retry.
FR-7.7	The system shall confirm payroll completion to the Manager/Admin with a summary report once processing finishes.

3.2.8 Reporting

ID	Requirement
FR-8.1	The system shall allow a Manager/Admin to view sales and stock reports.
FR-8.2	The system shall allow a Manager/Admin to view demand forecast results as part of, or alongside, stock reports.

3.3 Use-Case-Driven Requirement: Process Mobile Money Payment

This requirement is elaborated separately because it is central to the system's reliability and auditability guarantees.

Field	Description
Primary actor	Cashier (on behalf of Customer)
Preconditions	Sale total has been computed; the customer has an active mobile money account.
Main flow	(1) Cashier selects mobile money as the payment method; (2) the system sends a payment request to the gateway; (3) the gateway prompts the customer for PIN confirmation; (4) the customer confirms; (5) the gateway returns approval; (6) the system updates stock and prints a receipt.
Alternate flow	If the gateway declines or times out, the Cashier is prompted to retry or select another payment method.
Postconditions	The sale is recorded, stock is decremented, and payment is confirmed.

Chapter 4

Non-Functional Requirements

4.1 Reliability and Consistency

NFR-1 Mobile money payment confirmation shall be received before a sale is marked as paid.

NFR-2 Stock decrements shall occur only after payment approval, to avoid an inconsistent inventory state in the event of payment failure.

4.2 Scalability

NFR-3 The Prediction Engine shall be a separate logical component so that forecast computation (batch or scheduled runs) does not block point-of-sale transactions.

4.3 Security

NFR-4 Staff authentication shall reuse a common credential model across all staff-facing roles rather than duplicating credential handling per role.

NFR-5 Mobile money references and payroll data shall be treated as sensitive information and access-controlled to Manager/Admin roles only.

4.4 Auditability

NFR-6 Every sale shall be linked to a payment record carrying a mobile money (or transaction) reference.

NFR-7 Every payroll run shall produce a report that enables reconciliation against the mobile money provider's statements.

4.5 Usability

NFR-8 Cashier-facing checkout flows shall clearly present the current sale total and selected payment method before payment is initiated.

NFR-9 Failed or timed-out payments shall present the Cashier with a clear retry-or-switch-method option rather than a silent failure.

Chapter 5

Data Requirements

5.1 Core Domain Entities

The following entities, derived from the system's domain model, define the minimum data the system must persist to satisfy the functional requirements in Chapter 3.

Entity	**Required Attributes (minimum)**
Account	Account ID, username, password hash, role
Customer	Customer ID, name, phone, mobile money account, loyalty points
Staff	Staff ID, name, role, base salary, mobile money account
Product	Product ID, name, category, unit price, reorder level, quantity in stock
Supplier	Supplier ID, name, contact, lead time (days)
InventoryItem	Batch ID, quantity, expiry date, date received
Sale	Sale ID, date/time, total amount, status
SaleItem	Quantity, unit price, subtotal, linked product
Payment	Payment ID, amount, status, timestamp, confirm/refund capability
MobileMoneyPayment	Mobile money reference, phone number, provider
OnlinePayment	Transaction reference, gateway name, card last 4 digits
DemandForecast	Forecast ID, period, predicted quantity, trend
Payroll	Payroll ID, period, gross pay, deductions, net pay

5.2 Data Retention and Integrity

- Payment and payroll records shall be retained in a form suitable for reconciliation against mobile money provider statements (see NFR-6, NFR-7).
- Each inventory batch's expiry date shall be tracked to support expiry-based business rules (e.g., excluding expired batches from sale).

Chapter 6

Traceability and Closing Notes

6.1 Traceability to the Software Design Document

Each functional requirement in Chapter 3 traces to a corresponding element of the SmartStock Software Design Document (v1.0): the four functional areas map to Sections 1.2 and 2.2 of the SDD; account/checkout/payment requirements map to the SDD's use case diagram (Section 3.1) and the "Process Mobile Money Payment" use case specification (Section 3.3); inventory, demand prediction, and payroll requirements map to the class diagram (Chapter 4) and the three sequence diagrams and activity/state diagrams in Chapter 5 of the SDD; non-functional requirements map directly to Chapter 6 of the SDD.

6.2 Open Items for the Group

- Complete the missing matricules for Kamdeu Yamdjeuson and Neil Marshall.
- Confirm whether Online Payment (card) processing is in scope for the current iteration, or mobile money only.
- Define specific, measurable performance targets (e.g., maximum payment gateway timeout duration, maximum acceptable forecast generation time) once early testing data is available.